

Liam Tanner

liamtanner.dev | liam.tanner@brentwood.ca | linkedin.com/liam-tanner | github.com/liamtannerr

TECHNICAL SKILLS

Languages: Python, C++, C, SQL, TypeScript

Backend & Systems: FastAPI, PostgreSQL, MongoDB, RESTful APIs, ETL Pipelines, ROS 2, CUDA, Linux

AI & Machine Learning: PyTorch, CNNs, Optimization Math, LLM Orchestration, RAG, Scikit-learn

Web & Tools: React, Docker, Git, Agile Methodologies, AWS (Cloud Practitioner Certified)

EDUCATION

University of Victoria

Victoria, BC

Bachelor of Software Engineering

Sep 2021 to Aug 2026

- 4.00/4.00 GPA - 93% Avg (2025/2026)
- Coursework: Optimization for ML, Computer Vision, Data Mining, Operating Systems, Databases, Networks

EXPERIENCE

Software Developer

Jun 2025 to Dec 2025

Natural Resources Canada

Victoria, BC

- Built automated ETL pipelines alongside a team of software engineers to transform PostgreSQL datasets into SQLite and flatfile releases, reducing processing time from hours to seconds.
- Developed a release-versioning system using SQL and encryption to encode and track metadata.
- Wrote Linux Bash scripts to extract and repair a legacy website lacking source code, allowing the team to spin up a fully functional local environment with a single command.

Software Engineer

Jan 2025 to Aug 2026

AUVIC - Autonomous Underwater Vehicles Club

Victoria, BC

- Built a ROS 2 dynamic configuration pipeline that reads PID and Kalman filter values from YAML files, eliminating C++ recompilation downtime and enabling real-time tuning during pool testing.
- Resolved thruster saturation limits and built a state machine for autonomous navigation, advancing the team to the semi-finals at RoboSub 2026.

Research Assistant - Induced Seismicity Project

Sep 2023 to Apr 2024

Natural Resources Canada

Sidney, BC

- Built 3D CO2 injection simulations using Python meshes and the OpenGeoSys finite element solver, successfully implementing the workflow for the first time at the facility.
- Modeled a 3-layer finite element simulation of the SECURE INGA disposal well utilizing real-world depth profiles and material properties.

PROJECTS

AI Incident Orchestrator | *Python, PostgreSQL, LLMs, RAG*

May 2026 to Aug 2026

- Built an AI orchestration pipeline using Python, PostgreSQL, and the Groq API to automatically diagnose cybersecurity threats and output mitigation steps.
- Leveraged LLMs and RAG context pipelines to evaluate unstructured incident data and generate recommendations in a distributed environment.

EfficientAD Anomaly Detector | *Python, PyTorch, Computer Vision*

Mar 2026

- Implemented the EfficientAD visual anomaly detection pipeline in PyTorch, removing bottlenecks from the student training algorithm to process 70,000 iterations in under 2 hours.
- Achieved an AUROC score of 0.88 on an industrial dataset with sub-millisecond inference latency.

Convolutional Engine | *C++, CUDA, Hardware Optimization*

Apr 2026

- Wrote a custom image processing library in C++ and CUDA optimized for parallel GPU execution.
- Implemented shared memory tiling to bypass global memory bottlenecks and maximize GPU throughput for convolutional computations.

Virtual Triage System | *React, TypeScript, Python, FastAPI, MongoDB*

Sep 2024 to Nov 2024

- Developed a virtual triage web application using ReactJS, Python, and FastAPI to queue and assign priorities to prospective emergency department patients.
- Integrated MongoDB to store login information and patient triage data, enabling remote nurses to assign priorities and broadcast real-time wait times.